INTERIM FEDERAL SPECIFICATION

STAIN, LATEX, EXTERIOR FOR WOOD SURFACES

This Interim Federal Specification was developed by the U. S. Department of Commerce, National Bureau of Standards, Washington, DC 20234, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers latex stains for wood surfaces.

1.2 Classification. The stains specified herein shall be designed and furnished in the following classes (see 6.2):

Class A - Semi-transparent
Class B - Opaque

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification.

PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials: Packaging, Packing and Marking.

Federal Standards:

Fed. Test Method Std. No. 141/GEN - Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling, and Testing.
Fed. Test Method No. 141/1011 - Inspection (General).
Fed. Test Method No. 141/1021 - Sampling (General).
Fed. Test Method No. 141/1031 - Contractor Inspection Responsibility for FSC Group 80 Items (Other than Brushes, 8020, and Adhesives, 8040).
Fed. Test Method No. 141/2031 - Preparation of Wood Panels.
Fed. Test Method No. 141/3011 - Condition in Container.
Fed. Test Method No. 141/4250 - Color of Pigmented Coatings.
Fed. Test Method No. 141/4321 - Brushing Properties.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 562 - Consistency of Paints Using the Stormer Viscosimeter.
D 659 - Resistance to Chalking of Exterior Paints. Evaluating Degree of.
D 822 - Light-and-Water-Exposure Apparatus (Carbon-Arc Type) for Testing Paint, Varnish, Lacquer, and Related Products. Recommended Practice for Operating.
D 1640 - Drying, Curing, or Film Formation of Organic Coatings at Room Temperature.
D 2244 - Instrumental Evaluation of Color Differences of Opaque Materials.
D 2247 - Coated Metal Specimens at 100 Percent Relative Humidity.
D 3273 - Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber.
D 3274 - Surface Disfigurement of Paint Film by Fungal Growth or Soil and Dirt Accumulation.
D 3335 - Low Concentration of Lead in Paint by Atomic Absorption Spectroscopy.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The ready-mixed stain as received shall consist of the pigment and vehicle specified, so combined as to produce a stain meeting this specification. The stain shall be free from materials which will be toxic to personnel under normal conditions of use.

3.1.1 Pigments. The pigments, including extenders, shall be lightfast and alkali resistant.

3.1.2 Vehicle. The vehicle shall consist of a water dispersion of acrylic, alkyd modified acrylic or vinyl acrylic resins. A preservative shall be present together with the minimum amounts of other necessary additives such as emulsifiers, pigment dispersants, antifoaming agents, etc. provided the stain meets the specification.

3.2 Quantitative requirements. The stain shall conform to the quantitative requirements of table I.

Table I. Quantitative requirements

Characteristics	Class A		Class B	
	Min.	Max.	Min.	Max.
Pigment, percent by weight of stain	4	15	20	25
Nonvolatile organic material, percent by weight of stain	10	25	12	20
Viscosity, Krebs-Stormer, shearing rate 200 r.p.m.				
Grams	75	150	175	250
Equivalent Krebs Units (K.U.)	55	70	75	90
Dry hard, hour	---	1	---	1
Lead, metal, percent by weight of nonvolatile matter	---	0.5	---	0.5

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3.3 Qualitative requirements.

3.3.1 Condition in container. The stain as received shall be ready-mixed, and shall show no evidence of mold growth, livering, skinning, putrefaction, rust from corrosion of the container, or hard settling of the pigment. Any settled pigment shall be readily dispersible in the liquid portion by hand stirring to form a smooth homogeneous stain, free from persistent foam.

3.3.2 Color. Unless otherwise specified (see 6.2) the color shall match that of the standard color chip in Fed. Std. No. 595, when tested as specified in 4.3.2.

3.3.3 Working properties. The stain shall be easily applied by brush, roller, or spray equipment. When tested as described in 4.3.3, the stain shall brush easily at package consistency and shall permit lapping without difficulty. When dry, the surface shall present a smooth, uniform appearance without streaking, conspicuous laps, or objectionable brush marks.

3.3.4 Storage stability. The stain, as received, shall withstand the freeze-thaw test and the heat storage test described in 4.3.4 with a viscosity change not greater than 4 K.U. After completion of these tests the stain applied to a wood panel shall dry to a smooth, uniform finish.

3.3.5 Stain resistance. When tested as described in 4.3.5, the stain shall show a color difference no greater than 4 units after exposure.

3.3.6 Accelerated weathering resistance. The stain coated panels shall be prepared and tested as described in 4.3.6. The stain shall show a chalk rating no lower than 8 and a color difference no greater than 4 units after exposure.

3.3.7 Fungus resistance. When tested as specified in table II, stain-coated panels shall have a disfigurement rating of 4 or greater as described in ASTM D 3274.

3.3.8 Shelf storage life. When stored as specified in 4.3.11, the stain shall meet the requirements of 3.3.1 and 3.3.2, and viscosity and dry hard time shall vary not more than plus or minus 10 percent from the values specified in table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 The supplier shall certify to the contracting officer that his stain formulation meets the requirements of 3.3.8. When certificates of compliance are submitted, the Government reserves the right to check such items to determine the validity of the certificate.

4.2 Quality conformance inspection.

4.2.1 Sampling and inspection. Sampling and inspection shall be in accordance with methods 1011, 1021, and 1031 of Fed. Test Method Std. No. 141, as applicable.

4.3 Tests. Samples selected in accordance with 4.2.1 shall be tested as specified in table II. Nonconformance to the requirements of the applicable test methods or failure to pass any test shall be cause for rejection of the lot which the sample represents.

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TABLE II. Index

Characteristics	Requirement Reference	Test Method	Paragraph Reference
Condition in container	3.3.1	3011 1/	4.3.1
Color	3.3.2	4250 1/	4.3.2
Working properties	3.3.3	4321 1/	4.3.3
Storage stability	3.3.4		4.3.4
Stain resistance	3.3.5		4.3.5
Accelerated weathering resistance	3.3.6	D 822 2/	4.3.6
Fungus resistance	3.3.7	D 3273 2/ D 3274 2/	4.3.7
Pigment	Table I		4.3.8
Nonvolatile organic material	Table I		4.3.8
Viscosity	Table I	D 562 2/	
Dry hard time	Table I	D 1640 2/	
Lead content	Table I	D 3335 2/	4.3.9
Identification of latex resin	3.1.2		4.3.10

1/ Fed. Test Method Std. No. 141.
2/ ASTM.

4.3.1 Condition in container. Examine the stain as received in accordance with method 3011 of Fed. Test Method Std. No. 141 for compliance with 3.3.1.

4.3.2 Color. The stain shall be applied to a clean, smooth plate-glass panel with a 0.003 inch (approximately 0.006-inch gap clearance) [0.0762 mm] Bird film applicator or similar blade which produces the same film thickness. Allow film to dry for 24 hours. Compare the color of the dried film in accordance with method 4250 of Fed. Test Method Std. No. 141.

4.3.3 Working properties. Determine the brushing properties in accordance with method 4321 of Fed. Test Method Std. No. 141. For the substrate, use western red cedar panels of approximately 152 by 457 mm (6 by 18 inches) prepared in accordance with method 2031 of Fed. Test Method Std. No. 141. Evaluate for compliance with 3.3.3.

4.3.4 Storage stability.

4.3.4.1 Specimens and exposure. Fill two resin-lined friction-top cans with the sample and close the cans tightly. Expose one can and contents three times to the following freeze-thaw cycle:

(a) Low temperature of -9 ± 2°C (+15 ± 3°F) for 16 hours.
(b) High temperature of 25 ± 3°C (77 ± 5°F) for 8 hours.

Store the other can and contents in an oven at 49°C (120°F) for 1 week.

4.3.4.2 Procedure:

(a) At the completion of the exposure test, measure the viscosity (ASTM D 562) of both specimens and compare with the original viscosity to determine compliance with 3.3.4.
(b) Brush the stain on a wood panel and observe while brushing and after drying whether the stain is normal and usable in all respects. Compare the unexposed stain with respect to freedom from coagulation, agglomeration, and change in color.

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4.3.5 Stain resistance. Prepare western red cedar panels of approximately 75 by 200 mm (3 by 8 inches) in accordance with method 2031 of Fed. Test Method Std. No. 141. Draw down a 51 mm (2 inch) wide film of the stain to 0.003 inch (0.0762 mm) dry film thickness and allow to cure for 7 days. Measure the color according to ASTM D 2244. Place the panels, coated face down, in a humidity cabinet 1/ as described in ASTM D 2247. Use a continuous condensation cycle and a 38 ± 1°C air temperature for 3 days as the test exposure condition. Remove the panels and allow to dry for 24 hours. Measure the color difference of the exposed panels. Observe for compliance with 3.3.5.

4.3.6 Accelerated weathering resistance. Prepare western red cedar panels approximately 75 by 200 mm (3 by 8 inches) in accordance with method 2031 of Fed. Test Method Std. No. 141. Draw down a 51 mm (2 inch) wide film of the stain to 0.003 inch (0.0762 mm) dry film thickness and allow to cure for 7 days. Measure the color according to ASTM D 2244. Expose the panels for 300 hours to accelerated weathering in accordance with method ASTM D 822. Remove the panels and allow to dry for 24 hours. Measure the color difference of the exposed panels. Measure the chalking in accordance with ASTM D 659. Observe for compliance with 3.3.6.

4.3.7 Fungus resistance. Test the stain in accordance with ASTM D 3273 and D 3274 for compliance with 3.3.7.

4.3.8 Pigment and nonvolatile organic material. Spread thinly approximately 5 grams of a well-mixed stain sample over the surface of a tared petri dish of borosilicate glass, approximately 10 centimeters in diameter, and obtain the exact total weight. Dry in a vented regular oven at 105°C to constant weight. (This requires overnight drying.) After obtaining the weight of the total nonvolatile material, ash in a muffle furnace, raising the temperature gradually to 450-500°C. Allow 2 hours for complete combustion of the organic material. The furnace door should be opened carefully when removing the sample, as the remaining pigment is very light and easily scattered. Place the sample in a desiccator, and cool to room temperature. Obtain the weight of the ashed material.

4.3.8.1 Calculations.

Weight of total nonvolatile material - weight of ashed material = weight of nonvolatile organic material.

$$\frac{\text{Weight of ash} \times 100}{\text{total weight of sample}} = \text{pigment, percent by weight of stain}$$

$$\frac{\text{Weight of nonvolatile organic material} \times 100}{\text{total weight of sample}} = \text{nonvolatile organic material, percent by weight of stain}$$

4.3.9 Lead content. Determine the lead content of the stain in accordance with ASTM D 3335 for compliance with table I.

4.3.10 Identification of latex resin. Weigh 15 grams of a well-mixed stain sample into a 100 ml lipped centrifuge tube. Add a small amount of hot acetone and stir until homogeneous. Dilute to 80 to 85 ml with hot acetone, and centrifuge at 1500 r.p.m. for 30 minutes. Decant and evaporate the decantate to 30 ml, and then cool.2/ If the resin precipitates from solution, filter, dissolve the precipitate in acetone, and produce a film on a sodium chloride disc. Vacuum dry the coated disc, and obtain an infrared spectrum. The film thickness shall be such that the absorbance at 5.8μ is 1 or more.

If no resin precipitates from the cooled 30 ml decantate, add the decantate while stirring to 60 ml of n-hexane. Allow the precipitate to settle and decant the supernatant. Dissolve the precipitate in acetone and proceed as described above.

1/ Available from the Q-Panel Company, Cleveland, Ohio 44135.
2/ CAUTION: Vapors of acetone are highly flammable. Condense this solution on a steam bath.

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Compare the spectrum obtained with published reference spectra. The presence of an oil or alkyd modification is indicated by an increased absorbance at 3.4μ, a broad absorbance band at 2.8u, and possible absorbance at 6.25μ.

4.3.11 Shelf storage life. In a one-pint glass jar with a close fitting top, store a pint of the stain for 1 year from the date of manufacture in a relatively undisturbed condition at 25 ± 1°C (77 ± 2°F). Test for compliance with 3.3.8 (see 4.1.1).

4.3.12 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected to determine conformance to section 5 in accordance with PPP-P-1892.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The stain shall be packaged, packed, and marked in accordance with PPP-P-1892. Packaging shall be level A, B, or C and packing shall be level A, B, or C, as specified (see 6.2). The stain shall be furnished in the size of container specified (see 6.2).

5.1.1 Special marking. In addition to marking specified by 5.1, each unit container, intermediate container, and shipping container shall be marked as follows.

(a) Date of manufacture by month and year (not in code).
(b) Date of first reinspection (insert date one year after date of manufacture).
(c) "Store above 0°C (32°F)."

6. NOTES

6.1 Intended use. This stain is intended for new or previously stained exterior wood surfaces. Do not apply over previously painted surfaces. Clean surfaces thoroughly to remove all loose material, foreign matter, and mildew before application of this stain. Avoid application in direct sunlight, in rainy or threatening weather, or when the ambient temperature is below 10°C (50°F). If two-coat application is necessary, allow 2 hours drying time before application of the second coat.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Color required (see 3.3.2).
(d) Level of packaging and level of packing required (see 5.1).
(e) Special marking required (see 5.1.1).
(f) Size of container required (see 5.1).

6.3 Basis of purchase. The stain should be purchased by volume, the unit being 1 gallon (equal to 3.785 liters or 231 cubic inches). The temperature of the stain should not be in excess of 20°C (68°F) at time of volume measurement.

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